

| <b>S.No</b> | <b>Architecture Component / Layer</b> | <b>Technology Chosen</b>                                                                                                                                           | <b>Justification / Purpose</b>                                                                                                                                                                                                                                                                    |
|-------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>    | <b>Frontend / Client-Side</b>         | <i>ServiceNow Incident form (Platform UI / Next Experience), Client Scripts and UI Policies written in client-side JavaScript using the g_form and g_user APIs</i> | <i>Client Scripts (onLoad, onChange, onSubmit) and UI Policies run in the browser, so field validation, mandatory/read-only/hidden behaviour and on-screen messages respond instantly without a page reload. The platform form is responsive and reuses the same logic across the form views.</i> |

|   |                              |                                                                                                                                                                       |                                                                                                                                                                                                                    |
|---|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | <b>Backend / Server-Side</b> | <i>ServiceNow platform: Incident table, Business Rules, ACLs and Script Includes (server-side JavaScript), with GlideAjax for asynchronous client-to-server calls</i> | <i>Server-side rules and ACLs re-validate what the client scripts enforce so the checks cannot be bypassed. Script Includes keep reusable logic in one place, and the built-in platform speeds up development.</i> |
|---|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Technology Stack

|                      |                                                |
|----------------------|------------------------------------------------|
| <b>Date</b>          | 30 September 2026                              |
| <b>Team ID</b>       | SWTID-2026-5283                                |
| <b>Project Name</b>  | Implement Client Script & UI Policy (Incident) |
| <b>Maximum Marks</b> | 4 Marks                                        |

#### Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

#### Technology Stack Details

| S.No | Architecture Component / Layer      | Technology Chosen                                                                                                                           | Justification / Purpose                                                                                                                                                                                                           |
|------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | <b>Database / Data Storage</b>      | <i>ServiceNow platform database (relational; Incident [incident] table with its reference tables such as sys_user and assignment group)</i> | <i>Incident data, dictionary entries and script records are stored in the managed relational database, so no separate database setup is needed. It provides audit history, field-level security and reporting out of the box.</i> |
| 4    | <b>Cloud / Hosting / Deployment</b> | <i>ServiceNow Personal Developer Instance (PDI), hosted in the ServiceNow cloud</i>                                                         | <i>A free, cloud-hosted instance that is quick to provision and needs no servers to maintain. Changes are captured in Update Sets so they can be promoted to test and production instances.</i>                                   |
| 5    | <b>Version Control &amp; CI/CD</b>  | <i>ServiceNow Update Sets and Studio source control, with GitHub for storing the script source and documentation</i>                        | <i>Update Sets track every Client Script and UI Policy change and make it repeatable across instances. GitHub gives a backup and version history for the team, and the ATF (Automated Test Framework) can run the form tests.</i> |

|          |                                       |                                                                                                                                                                            |                                                                                                                                                                                                                 |
|----------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6</b> | <b>Third-Party APIs / Other Tools</b> | <i>ServiceNow Script Debugger and JavaScript log/console, Chrome/Edge browser developer tools, ServiceNow Automated Test Framework (ATF); no external third-party APIs</i> | <i>Used to debug and test client-side scripts, check for console errors and confirm each UI Policy and Client Script behaves correctly on the Incident form. The project needs no external API integration.</i> |
|----------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|